

AMGA, Russian Federation

WMO: 249620

Lat: 60.900N

Lon: 131.983E

Elev: 146

StdP: 99.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -52.0	(c) -50.3	(d) -53.4	(e) 0.0	(f) -51.7	(g) -51.7	(h) 0.0	(i) -50.1	(j) 5.2	(k) -30.0	(l) 4.6	(m) -30.0	(n) 0.2	(o) 210	(p) 0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 30.4	(d) 19.3	(e) 28.3	(f) 18.3	(g) 26.1	(h) 17.5	(i) 21.0	(j) 27.1	(k) 19.8	(l) 25.9	(m) 18.6	(n) 24.2	(o) 2.7	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.0	(b) 14.0	(c) 23.5	(d) 17.7	(e) 12.9	(f) 22.2	(g) 16.5	(h) 11.9	(i) 21.2	(j) 61.7	(k) 27.4	(l) 57.6	(m) 26.0	(n) 53.7	(o) 24.3	(p) 25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.1	5.2	-53.2	33.1	2.5	2.2	-55.0	34.7	-56.5	35.9	-57.9	37.2	-59.7	38.8
			-53.0	23.0	2.2	1.6	-54.5	24.1	-55.8	25.0	-57.0	25.9	-58.6	27.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.4	-39.6	-36.1	-21.2	-4.9	7.4	15.4	18.6	15.0	6.0	-7.7	-27.7	-39.3
	DBStd	22.30	6.64	6.27	7.70	6.22	4.02	3.86	3.60	3.76	4.33	6.98	9.30	7.04
	HDD10.0	7694	1538	1291	968	447	100	4	0	5	133	549	1131	1528
	HDD18.3	10187	1796	1524	1227	697	338	102	41	117	371	808	1381	1786
	CDD10.0	626	0	0	0	0	20	167	267	159	12	0	0	0
	CDD18.3	76	0	0	0	0	0	14	49	12	0	0	0	0
	CDH23.3	1188	0	0	0	0	8	313	673	192	1	0	0	0
	CDH26.7	369	0	0	0	0	1	81	242	46	0	0	0	0
Wind	WSAvg	1.8	0.8	0.8	1.5	2.3	2.9	2.5	2.1	2.1	2.2	2.0	1.3	0.8
Precipitation	PrecAvg	282	10	7	6	8	23	37	48	55	40	22	17	9
	PrecMax	405	20	17	20	29	71	91	122	200	183	57	42	17
	PrecMin	194	3	2	0	0	4	9	13	1	8	4	5	2
	PrecStd	65	4	4	4	8	15	19	28	44	34	13	9	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.0	-13.3	2.9	13.9	24.5	31.2	33.9	30.7	21.9	7.5	-1.7	-14.6
		MCWB	-17.4	-14.0	-0.2	6.2	13.7	18.2	20.9	19.9	13.9	4.5	-2.3	-14.8
	2%	DB	-23.0	-19.4	-1.0	10.1	21.0	29.0	31.5	27.6	18.2	4.3	-4.9	-22.0
		MCWB	-23.3	-19.7	-3.4	3.8	11.6	17.9	20.0	18.6	12.3	2.0	-5.4	-22.3
	5%	DB	-26.5	-23.4	-4.6	7.2	18.5	27.0	29.5	25.3	15.8	2.3	-9.6	-25.6
		MCWB	-26.8	-23.7	-6.3	2.3	10.2	16.8	19.3	17.3	11.1	0.6	-10.0	-25.8
	10%	DB	-29.5	-26.4	-8.0	4.8	16.1	24.9	27.3	23.2	13.4	0.6	-14.0	-29.1
		MCWB	-29.6	-26.6	-9.2	1.0	8.9	16.0	18.6	16.3	9.7	-0.5	-14.3	-29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.3	-13.8	-0.3	6.3	14.6	20.7	23.0	21.6	14.7	5.0	-2.2	-14.8
		MCDB	-16.9	-13.3	2.8	12.6	22.8	27.9	29.4	26.3	19.6	7.3	-1.8	-14.6
	2%	WB	-23.2	-19.6	-3.3	4.3	12.3	19.0	21.7	20.0	13.3	2.4	-5.4	-22.3
		MCDB	-23.0	-19.4	-1.0	9.4	19.5	25.8	28.3	25.4	16.9	3.9	-5.0	-22.1
	5%	WB	-26.7	-23.7	-6.3	2.6	10.6	17.9	20.6	18.6	11.8	0.9	-9.9	-25.8
		MCDB	-26.5	-23.4	-4.6	6.6	16.8	24.4	27.0	23.8	14.7	2.0	-9.6	-25.7
	10%	WB	-29.7	-26.6	-9.3	1.2	9.4	16.8	19.6	17.1	10.2	-0.5	-14.3	-29.1
		MCDB	-29.6	-26.4	-8.0	4.4	15.0	23.5	25.6	21.8	12.9	0.6	-14.0	-29.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	10.6	17.4	15.0	12.1	13.9	12.8	12.4	10.4	8.2	8.2	6.5
		MCDBR	11.1	13.4	18.7	14.7	17.5	18.6	17.8	16.8	14.9	8.8	9.9	10.5
		MCWBR	11.0	13.1	16.3	9.3	9.3	8.7	7.8	8.3	9.0	6.4	9.7	10.4
	5% WB	MCDBR	11.2	13.3	18.5	13.8	15.0	15.9	14.4	14.2	12.4	7.8	10.0	10.5
		MCWBR	11.0	13.0	16.2	9.1	8.2	8.0	7.1	7.6	8.0	5.9	9.7	10.5
Clear-Sky Solar Irradiance	taub	0.223	0.265	0.306	0.371	0.466	0.419	0.507	0.448	0.353	0.295	0.232	0.193	
	taud	1.993	2.065	2.051	2.008	1.986	2.257	1.983	2.127	2.369	2.275	2.006	1.906	
	Ebn at Noon	504	707	798	807	752	809	712	727	758	675	474	365	
	Edh at Noon	43	76	110	141	157	122	157	126	82	62	41	28	
All-Sky Solar Radiation	RadAvg	0.46	1.50	3.19	5.00	5.23	5.90	5.42	4.21	2.64	1.25	0.60	0.23	
	RadStd	0.02	0.06	0.12	0.18	0.46	0.37	0.33	0.32	0.22	0.13	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.64	+2.06	+1.58	N/A	N/A	N/A	-201	-178	N/A	N/A
Regional (6 neighbors)	+0.65	+1.19	+1.05	N/A	N/A	N/A	-220	-232	N/A	N/A

Nomenclature: See separate page